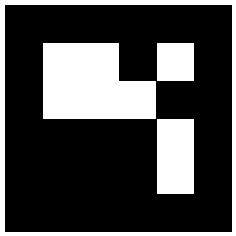
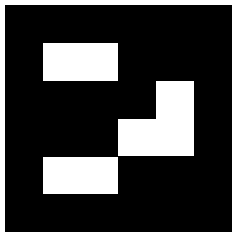
id=2



id=3



Robot Workshop for Beginners

Build and program a small line-following robot in one day!

Saturday, December 5, 2026

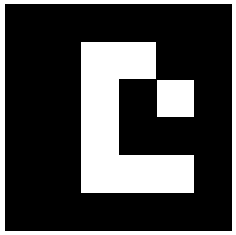
1:00 p.m. – 5:00 p.m.

Room 204, Engineering Building

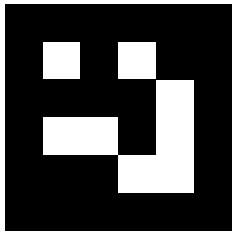
- Fee: $15 (parts included, you keep the robot)
- Capacity: 20 people (first come, first served)
- Age: 12 and up; no experience needed

Register by November 30 at robots@example.edu

id=1



id=4



cut here

Print scale check

Measure the bar below. It must be exactly 100 mm. If not, printer scaling is on.

Specification

dictionary : 4X4_50

paper orientation : the RIGHT edge of this sheet is the TOP

marker ids : 2, 3, 4, 1 (clockwise from top-left, in that orientation)

black square side : 30.0 mm <- use this for the marker_size parameter

quiet zone : 7.5 mm (white, all four sides)

paper : 210 x 148.5 mm (A4 halved)

marker inset : 8.0 mm from paper edge

min camera dist. : 0.151 m (recommend 0.18 m or more)

camera fov assumed: 69.4 x 42.5 deg (D435 color, nominal)

generated : 2026-11-02

Camera framing (independent of the reading orientation above): let the 210 mm side of the paper run across the WIDE axis of the image. Across the narrow axis it needs 0.28 m instead.